

Evaluation Overview

The project should be evaluated with chat-only, document-grounded, web-grounded, and mixed document-web questions.

Routing accuracy measures whether the router selected CHAT, RAG, WEB_SEARCH, or RAG_WEB correctly.

Retrieval quality can be measured with Recall@5, MRR, and nDCG@5.

Answer Quality Metrics

Faithfulness measures whether the answer is supported by retrieved evidence.

Citation accuracy measures whether cited sources actually contain the facts being claimed.

Answer completeness measures whether the response covers the important parts of the question.

Latency measures the trade-off introduced by reranking and verification.

Conclusion

A strong AI engineering portfolio project should not only work interactively, but should also explain how decisions are made.

The strongest evidence is a comparison showing that hybrid retrieval performs better than semantic-only or keyword-only retrieval.

The final system should be controllable, measurable, and grounded in explicit sources.